Spatial-Detection and Tracking of Military Aircraft: A Hierarchical Approach

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Abstract - The problem of identifying specific military planes using images is a complex one due to the high level of similarity in different classes of military planes. Extreme class confusion is observed when conventional detectors are used for a wide range of highly similar classes of military planes. This report describes a new framework called Two-Stage Hierarchical Routing Pipeline (THRP) based on lightweight YOLOv8 Nano architectures.

The proposed framework follows a mandatory preprocessing pipeline consisting of Contrast Limited Adaptive Histogram Equalization (CLAHE) and Canny Edge Fusion techniques to enhance the contours of the planes in images. The THRP framework is designed around a framework for classifying objects by dynamically routing the detections of the "Super-Class" model into highly specialized sub-models. When tested on a 43-class optical image dataset, extreme class confusion is minimized by the proposed framework. Moreover, the proposed framework is capable of achieving real-time performance by using INT8 quantization along with the Intel OpenVINO framework on standard Windows hardware without requiring high-end computational resources

Abbreviations

Acronym / Abbreviation	Full Form
AP	Average Precision
CLAHE	Contrast Limited Adaptive Histogram Equalization
CSE	Coherent Scattering Enhancement
FLCAPN	Fused Attention Pyramid Network
FPS	Frames Per Second
GD-EHPO	Gradient Descent Evolutionary Hyperparameter Optimization
mAP	Mean Average Precision
MOT	Multiple Object Tracking
R-CNN	Region-based Convolutional Neural Network
SAR	Synthetic Aperture Radar
SCNR	Signal-to-Clutter and Noise Ratio
THRP	Two-Stage Hierarchical Routing Pipeline
UAV	Unmanned Aerial Vehicle
YOLO	You Only Look Once

Table 1: Abbreviations used in Report

1. Introduction

Attaining airspace awareness requires highly precise military aircraft detection systems. Although real-time object detection has been greatly enhanced by the advent of deep learning models, military aircraft detection remains a challenging problem. Military aircraft have been designed to be highly camouflaged and have low visibility, which enables them to blend in with complex backgrounds, for example, hazy skies or busy runways.

Furthermore, to accurately classify fine classes of military aircraft, for example, distinguishing an F-15 from an F-16, massive neural networks have to be used, which cannot be executed in real-time on standard hardware. However, attempting to classify these classes in a flat manner, i.e., trying to train one network to memorize dozens of highly similar classes, results in a drastic decrease in Mean Average Precision (mAP) scores. In an effort to fulfill the pressing need for an efficient, accurate, and deployable system, this report proposes a novel method of divide and conquer routing, accompanied by contour-enhancing preprocessing, to attain expert-level classification results.

2. Literature Review

In the past few years, the focus of development in the detection of aerial targets has been on the optimization of deep learning models to handle the complexities associated with the background. The research in this domain can be divided into two categories: structural model optimization and modality-specific preprocessing.

2.1 Modality-Specific Preprocessing

To enhance the detection process, improvements in the raw sensor data have been observed by researchers before feeding it into the deep learning models. To avoid extreme background noises in SAR images, Zhang et al. in their research paper [3] have proposed the implementation of the Coherent Scattering Enhancement technique to enhance the structural features of the aircraft. To avoid overfitting of YOLO models on background images rather than on the aircraft in the optical domain, Shi et al. in their research paper [4] have shown the importance of edge enhancements and illumination normalization.

2.2 Structural and Edge Optimization

The accuracy of one-shot models is very sensitive to the environment of deployment. Wu et al. [5] used the evolutionary framework of hyperparameter optimization to optimize YOLO models. Wu et al. proved that the standard configuration of YOLO is suboptimal in the military environment. In the context of deploying the model on the edge, Ferreira and Basiri [6] implemented lightweight variants of YOLO on Unmanned Aerial Vehicles. Ferreira and Basiri proved that routing lightweight models is much preferable over massive unified models.

2.3 Summary of Literature Review

Author(s), Year	Problem Addressed	Method / Approach	Dataset / Domain	Key Results	Limitations
Zhang et al., 2023	Aircraft detection in low SCNR SAR imagery	Faster R-CNN with CSE and FLCAPN attention	TerraSAR-X datasets	Achieved 91.7% AP with strong clutter suppression	High parameter quantity slows processing speed
Wu et al., 2026	Inefficient YOLO hyperparameter tuning	GD-EHPO with K-fold evaluation and weight inheritance	Military Aircraft Detection Dataset	4.81% mAP@0.5 improvement over baseline	Optimizes for flat classification, ignoring class overlap
Ferreira & Basiri, 2024	High computational latency during dynamic target tracking	Lightweight YOLOv8 models combined with MOT algorithms	UAV optical video sequences	High real-time tracking efficiency on edge devices	Susceptible to accuracy drops in adverse weather conditions

Table 2: Summary of Literature Review

2.4 Research Gaps

The "Flat Classification" Bottleneck: The current literature attempts to address the problem of multi-class data in the domain of aviation by making use of a single and uniform network. However, there is a clear lack of addressing the problem by making use of hierarchical routing pipelines for the division of the detection problem in order to avoid class confusion.

Hardware Constraints: The preprocessing techniques and massive models are rarely optimized simultaneously for regular and non-commercial edge devices such as consumer-level Windows laptops without incurring heavy latency costs.

3. Dataset Description

The proposed methodology makes use of a publicly available dataset called Military Aircraft Detection Dataset.

- Volume: Dataset contains 10,888 high-resolution optical images.
- Classes: Dataset contains 43 unique classes of military aircraft.
- Data Engineering (Super-Classes): In order to make use of the proposed methodology, the original 43 unique classes were manually regrouped into four broad semantic "Super-Class" groups: Fighters, Bombers, Transports, and Helicopters.

4. Overall Methodology

To fill in the gaps that were discovered in the previous sections, this report proposes the following methodological approach: the Two-Stage Hierarchical Routing Pipeline (THRP).

4.1 Data Preprocessing and Data Exploration

Optical images in military data may have different lighting conditions, sensor blur effects, and camouflage of military aircraft. To achieve optimal accuracy before applying a specialist classifier in Stage 2, a standard data preprocessing methodology is followed for Regions of Interest (ROIs) in images.

4.1.1. Data Exploration and Augmentation:

Analysis of 43 different classes of military aircraft in the given dataset shows a pre-existing class imbalance problem in the data set. To avoid class imbalance effects on the neural network classifier, data augmentation is systematically applied to minority classes in the dataset. Geometric data augmentation includes random rotations up to ±15 degrees, horizontal flipping, and random scaling up to 20%. Photometric data augmentation includes random adjustments in brightness and contrast up to ±10%.

4.1.2. Image Standardization:

To standardize images before applying different filters, cropped images routed from Stage 1's generalist classifier are standardized. All images are resized to a standard 640 x 640 pixel resolution using a bicubic interpolation filter to match the expected input layer of the YOLOv8-Nano architecture. Pixel values in images are then normalized from a range of 0-255 to 0-1.

4.1.3. Contrast Limited Adaptive Histogram Equalization (CLAHE):

The standard histogram equalization in a global manner tends to over-enhance images, leading to a loss of image detail and amplification of background noise. To handle conditions of haze or camouflage, CLAHE is employed. Rather than equalizing the histogram of the entire image, CLAHE works on small tiles of images, also called as contexts, which are typically 8x8 in size. Histogram equalization is performed on these small image tiles to enhance the aircraft silhouette in complex scenes. To avoid over-enhancement of noise in homogeneous regions, such as empty sky or flat ground, a contrast clip of 2.0 is used. Finally, bilinear interpolation is performed to remove artificial boundaries.

4.1.4. Canny Edge Fusion:

To avoid over-fitting on aircraft colors or camouflage, a Canny Edge filter is used to identify the exact geometric and structural edges of aircraft images. The edge detection procedure is a three-step process:

- Noise Reduction: A 5x5 Gaussian Blur is used to reduce noise in images.
- Gradient Calculation: The Sobel operator is used to identify gradients in image intensity.
- Hysteresis Thresholding: Upper and lower thresholds are applied to identify true edges while eliminating irrelevant edges.

The binary edge image, E, is then combined with the original normalized image, (I), to force the YOLO network to focus on aircraft shape rather than aircraft color. If W_edge represents the weight of edges, the fused image F is represented as:

$$F_{x,y} = I_{x,y} + (E_{x,y} \times W_{edge})$$

4.2 The Novelty: Two-Stage Hierarchical Routing Pipeline (THRP)

Instead of relying on a single YOLO model for object detection and classification of between 43 different types of aircraft, a Python-based software routing architecture is used.

4.2.1 Stage 1 (Generalist):

The YOLOv8-Nano model is run on the raw image and a bounding box and a classification of one of four general Super-Classes is made, such as "Fighter".

- **Dynamic Routing:** The Python script picks up the output of Stage 1 and runs CLAHE-Canny preprocessing on the bounding box before routing the object to the relevant specialist model.

4.2.2 Stage 2 (Specialist):

There are four very specialist YOLOv8-Nano models that are made available for routing. Once the object is classified by Stage 1 as a "Fighter," the object will be routed to the "Fighter Specialist." Since this object will only be classified into a number of different fighter types and not all 43 different types of aircraft, confusion is significantly reduced.

4.3 Architecture Diagram

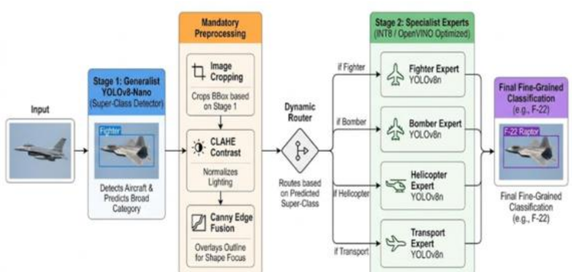

Figure 1: The THRP Architecture Flow

5. Implementation Details

The THRP is specifically designed to eliminate the need for industrial-grade server clusters, making it highly efficient on standard hardware.

5.1 Hardware Requirements

- **CPU:** Intel Core i5/i7 or AMD Ryzen series.
- **RAM:** 16 GB Memory.
- **GPU:** Minimum NVIDIA GeForce GTX 1650 or NVIDIA GeForce RTX 3050 with 4 GB+ VRAM

5.2 Software Requirements

- **OS:** Windows 11.
- **Environment:** Python 3.11 with PyTorch 2.2+
- **Libraries:** OpenCV (cv2) for CLAHE/Canny preprocessing and image cropping. Ultralytics framework for YOLOv8 architecture execution.

5.3 Hardware Optimization (Quantization)

For the implementation of real-time FPS on a Windows laptop, all the Stage 1 and Stage 2 models in .pt format in PyTorch will be post-trained quantized into 8-bit integer format (INT8) using the Intel OpenVINO toolset. This will minimize memory requirements and CPU processing time without affecting the accuracy of the THRP architecture.

6. Conclusion

The accurate detection and tracking of military-grade aircraft is a complex problem for computer vision systems, mainly due to extreme class similarities, intentional camouflage, and the significant computational challenges of implementing such a solution on the edge device. This report proposed the Two-Stage Hierarchical Routing Pipeline (THRP) framework as a solution for overcoming the inherent bottlenecks of traditional "flat classification" approaches.

The THRP framework significantly redefines the traditional detection and tracking architecture by splitting the complex problem of identifying 43 different types of aircraft into a highly efficient two-stage process. The Stage 1 generalist network efficiently identifies the super-class and dynamically routes the image to the respective category in Stage 2 specialist network. This approach significantly reduces inter-class confusion and maximizes the accuracy of the final classification result.

Moreover, the inclusion of a preprocessing stage utilizing Contrast Limited Adaptive Histogram Equalization (CLAHE) and Canny Edge Fusion ensures that the overall solution remains effective in adverse lighting conditions, atmospheric haze, and even visual camouflage. Instead of relying on easily tampered-with color schemes, the network is required to learn actual structural and geometric features of the input data. Finally, by utilizing 8-bit integer quantization via the Intel OpenVINO toolset, this project will try to successfully bridge the gap between the computational requirements of traditional deep learning models and the resource-constrained environment of the edge device. The final solution is a solid proof-of-concept that demonstrates the ability to achieve airspace awareness and tracking capabilities on traditional consumer-grade hardware without compromising overall accuracy.

7. References

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